

Solar India

648 inregistrari orare | 41 features

1. Incarcare date

- 2 fisiere CSV (Plant_1)
- Generare + Senzori meteo
- 22 invertoare agregate
- Format dd-mm-yyyy
- Perioada 15 mai - 17 iun 2020

2. EDA (notebook 03)

- Statistici descriptive
- Curba zilnica gauss
- Iradiere ↔ AC_POWER (>0.95)
- Productie max ~15.000 W
- Fara valori lipsa

3. Preprocesare (nb 04)

- Resampling 15min → orar
- Tratare NaN (interpolare)
- Features temporale + sin/cos
- Lag-uri 1, 2, 3, 24, 168
- Rolling mean+std (3, 24, 168)

4. Armonizare (nb 04b)

- Performance Ratio (Yield)
- dc_ac_ratio (eficienta inverter)
- Temperature derating (-0.4%/grad)
- Validari fizice (46 anomalii)
- Variabila season + virtual year

5. Output salvat

- data/processed/solar_india_features.parquet
- Shape: (648, 41)
- 0 NaN-uri
- Train/Val/Test cronologic
- Gata pentru ML

6. Urmatori pasi

- ML predictiv (Sesiunea 3)
- Optimizare orientare panouri
- LSTM probabil slab (set mic)
- RF + XGBoost favoriti
- SHAP pe eficienta solar



